Drone Communication System using STM32 and SX1280

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Introduction

With the increasing use of drones in surveillance, agriculture, delivery systems, and disaster management, reliable wireless communication has become an important requirement. Drone communication systems require low latency, secure data transfer, and stable long-range wireless links for proper operation.

This project focuses on developing a custom drone communication system using the STM32 microcontroller and the SX1280 RF transceiver module. The system receives control data from an RC transmitter using the CRSF protocol, processes the data using STM32, encrypts it using AES-128 encryption, and transmits it wirelessly over a 2.4 GHz RF link.

At the receiver side, the transmitted packets are received, decrypted, reconstructed into CRSF frames, and forwarded to the flight controller. The project combines embedded systems, RF communication, communication protocols, and wireless security to create a reliable drone communication link.

Objectives

- To study the STM32 embedded platform and peripheral interfacing.
- To interface the SX1280 RF transceiver with STM32 using SPI communication.
- To implement CRSF protocol communication for drone control signals.
- To develop a transmitter and receiver system for wireless drone communication.
- To implement AES-128 encryption for secure data transmission.
- To study frequency hopping techniques for improving communication reliability.

System Overview

The developed system consists of a transmitter and receiver architecture designed for secure drone communication. The transmitter receives RC channel data using the CRSF protocol, encrypts the data, and sends it wirelessly using the SX1280 RF transceiver.

The receiver continuously listens for RF packets, decrypts the received payload, reconstructs the channel information, and sends CRSF output to the flight controller.

System Flow

RC Transmitter → CRSF Protocol → STM32 (TX) → AES Encryption → SX1280 (TX) → RF Communication → SX1280 (RX) → AES Decryption → STM32 (RX) → CRSF Output → Flight Controller

Study of STM32 Platform

The STM32 microcontroller was used as the main processing unit of the communication system. STM32CubeIDE was used for firmware development, debugging, and peripheral configuration.

The following peripherals were studied and used during implementation:

- UART for CRSF communication
- SPI for SX1280 interfacing
- GPIO for interrupt and reset handling
- Timer-based operations for communication timing

The Hardware Abstraction Layer (HAL) library provided by STM32 was used to simplify peripheral control and firmware development. Initial testing involved UART transmission, SPI communication verification, and debugging using serial output.

Study of SX1280 RF Transceiver

The SX1280 is a 2.4 GHz RF transceiver designed for low-latency and high-speed wireless communication. It supports communication modes such as LoRa and FLRC, making it suitable for drone applications requiring reliable data transfer.

Important features of SX1280 include:

- Operation in 2.4 GHz ISM band
- Low communication latency
- High sensitivity receiver
- SPI-based control interface
- Support for packet-based RF communication

The module operates using command-based SPI communication, where commands are used to configure operating frequency, transmission mode, packet parameters, and interrupt handling.

Interfacing STM32 with SX1280

The SX1280 module was interfaced with STM32 using SPI communication. The following signals were used:

- NSS (Chip Select)
- SCK (SPI Clock)
- MOSI (Master Out Slave In)
- MISO (Master In Slave Out)
- BUSY pin
- RESET pin
- DIO1 interrupt pin

The initialization process included:

- Configuring SPI communication
- Resetting the transceiver module
- Setting operating frequency
- Configuring packet parameters
- Enabling interrupt handling

Proper handling of BUSY and DIO1 signals ensured stable communication between STM32 and SX1280.

CRSF Protocol Integration

CRSF (Crossfire) is a high-speed, low-latency serial communication protocol widely used in RC systems for transmitting control data between the receiver and flight controller. In this project, CRSF is used both at the transmitter side (input) and receiver side (output), making it a critical part of the system.

CRSF Frame Structure

A standard CRSF RC frame consists of:

- Header(Device Address)
- Length
- Frame Type:0x16forRCchannel data
- Payload
- CRC:Errordetection byte

CRSF Input Processing (Transmitter Side)

On the transmitter side, CRSF data is received via UART at high baud rates (~420000 baud). The STM32 processes the incoming data as follows:

- Byte-wise reception using UART polling
- Detection of valid header bytes (0xC8, 0xEA, 0xEE)
- Extraction of frame length
- Accumulation of full frame
- CRC verification using polynomial (0xD5)

Only valid frames with correct CRC are processed further. The payload is then unpacked to extract 16 channel values.

CRSF Output Generation (Receiver Side)
At the receiver, after RF reception and AES decryption, the channel values are reconstructed and sent to the flight controller using CRSF format.
Steps involved:
1. Convert microsecond values to CRSF raw format:
    a) Mapping range to 11-bit values (172 to 1811)
2. Pack channels into CRSF frame:
    a) 16channels packed into 22 bytes using bit manipulation
3. Construct full frame:
    a) Header =0xC8
    b) Length =0x18
    c) Type =0x16
    d) Payload
    e) CRC
4. Transmit frame via UART:
    a) USART1at420000baud

CRSF acts as the interface between: Human input (RC transmitter) , Embedded system (STM32) and the Flight controller
It ensures compatibility with standard drone control systems while maintaining low latency and high reliability.

Transmitter Design

The transmitter side performs the following operations:

- Receiving CRSF input data
- Extracting channel values
- Packing channel data into payload buffer
- Encrypting payload using AES-128
- Transmitting encrypted data through SX1280

The transmitter continuously monitors transmission status using interrupt flags to ensure successful packet transmission

Receiver Design

The receiver continuously listens for incoming RF packets using SX1280 receive mode.

Receiver operations include:

- Receiving RF packets
- Reading payload length
- Extracting RSSI and SNR values
- Decrypting received payload

- Reconstructing channel values
- Generating CRSF output

The processed data is then forwarded to the drone flight controller

AES Encryption Implementation

To improve communication security, AES-128 encryption was implemented in the system. AES operates on 16-byte data blocks using a 128-bit encryption key.

The encryption process includes multiple rounds consisting of:

- Sub Bytes
- Shift Rows
- Mix Columns
- AddRoundKey

In this project:

- The 32-byte payload was divided into two 16-byte blocks
- Each block was encrypted separately
- The receiver performed inverse AES operations for decryption

This ensured protection against unauthorized access and packet tampering during wireless transmission.

Frequency Hopping Study

A study on frequency hopping techniques was carried out to understand methods for improving communication reliability and resistance to interference.

Frequency hopping involves dynamically changing operating frequencies during communication in order to reduce the effect of noise, interference, and signal jamming. Initial software-level experimentation and configuration analysis were performed using the SX1280 platform.

Although complete implementation and testing were not achieved, the study provided useful understanding of advanced RF communication techniques used in modern drone systems.

Conclusion

A secure and reliable drone communication system was successfully developed using the STM32 microcontroller and SX1280 RF transceiver. The project involved the study and implementation of embedded systems, RF communication techniques, CRSF protocol integration, and AES-128 encryption for secure wireless transmission.

The developed system was capable of transmitting and receiving encrypted control data wirelessly with low latency and stable communication performance. The transmitter and receiver architecture was successfully implemented using SPI and UART communication interfaces, enabling efficient exchange of control information between the RC transmitter and the flight controller.

The project provided practical understanding of RF transceiver configuration, packet-based wireless communication, interrupt handling, and real-time embedded firmware development. Significant work was carried out in CRSF frame processing, payload handling, data packing and unpacking, UART communication at high baud rates, and synchronization between transmitter and receiver modules.

AES encryption was incorporated to improve communication security by protecting transmitted data from unauthorized access and tampering. The implementation of encryption and decryption operations also provided exposure to secure communication methods used in modern wireless systems.